ELECTRODYNAMICS II · SEMESTER 5

Electromagnetic waves

Module 3 · Answer key

40 worked answers. Each answer names the prepare topic to revise first. Mark splits are suggested study rubrics.

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Section A

Q1 Model · syllabus-based

Determine the direction and speed of $f(z,t) = A\cos(4z + 12t)$, with SI coordinates.

KNOW FIRST

T1 Travelling waves in one dimension P1 School physics bridge M1 Mathematics from the beginning

Keep $4z + 12t$ constant: $4\,dz/dt + 12 = 0$, so $dz/dt = -3$ m/s. A constant-phase point travels toward decreasing z.

ANSWER Speed 3 m/s toward −z.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q2 Model · syllabus-based

Write the one-dimensional wave equation and interpret its two derivatives.

KNOW FIRST

T1 Travelling waves in one dimension P1 School physics bridge M1 Mathematics from the beginning

$\partial_z^2 f = v^{-2}\partial_t^2 f$. The left side is spatial curvature; the right side is time acceleration divided by speed squared. It constrains how a disturbance propagates, rather than specifying one particular shape.

ANSWER $\partial_t^2 f = v^2 \partial_z^2 f$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q3 Model · syllabus-based

A vacuum wave has f=100 MHz. Find λ, k and ω.

KNOW FIRST

T2 Sinusoidal waves and phase P1 School physics bridge M1 Mathematics from the beginning

$\lambda = c/f = 3.00$ m; $k = 2\pi/\lambda = 2.094$ rad/m; $\omega = 2\pi f = 6.283 \times 10^8$ rad/s. MHz means 10^6 Hz.

ANSWER λ=3.00 m, k=2.094 rad/m, $\omega = 6.283 \times 10^8$ rad/s.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q4 Model · syllabus-based

Two points along a sinusoidal wave are separated by λ/4. What is their phase difference at the same time?

KNOW FIRST

T2 Sinusoidal waves and phase P1 School physics bridge M1 Mathematics from the beginning

$\Delta\phi = k\Delta z = (2\pi/\lambda)(\lambda/4) = \pi/2$ radians. They are a quarter-cycle apart; which leads depends on the ordered positions and propagation direction.

ANSWER Phase separation magnitude $\pi/2$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q5 Model · syllabus-based

Define polarization and explain why a vacuum plane EM wave can be polarized.

KNOW FIRST

T3 Polarization of a wave P1 School physics bridge M1 Mathematics from the beginning

Polarization specifies the motion of the E-field tip in the plane transverse to propagation at a fixed point. Two independent transverse components can have different amplitudes and phase relations, giving different polarization states.

ANSWER **Polarization concerns transverse E orientation and its time evolution.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q6 Model · syllabus-based

What amplitude and phase conditions produce circular polarization from two perpendicular components?

KNOW FIRST

T3 Polarization of a wave P1 School physics bridge M1 Mathematics from the beginning

The components have equal nonzero amplitudes and phase difference $\pm\pi/2$ modulo 2π. The magnitude of E is then constant while its tip rotates. Zero phase difference instead gives a straight line.

ANSWER **Equal perpendicular amplitudes in quadrature give a circle.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q7 Model · syllabus-based

Classify $E_x = 3\cos\phi$ and $E_y = 4\cos\phi$. Give peak magnitude and orientation.

KNOW FIRST

T3 Polarization of a wave P1 School physics bridge M1 Mathematics from the beginning

The component ratio is $E_y/E_x = 4/3$, constant whenever defined. The tip moves on a line at $\tan^{-1}(4/3) = 53.13°$ to +x. Peak magnitude is $\sqrt{3^2 + 4^2} = 5$ in the field units.

ANSWER **Linear polarization, peak 5, line angle 53.13°.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q8 Model · syllabus-based

Classify $E_x = 3\cos\phi$, $E_y = 4\sin\phi$ and write the tip's locus.

KNOW FIRST

T3 Polarization of a wave P1 School physics bridge M1 Mathematics from the beginning

Eliminate phase using $\cos^2\phi + \sin^2\phi = 1$. This gives $E_x^2/9 + E_y^2/16 = 1$, an ellipse with semi-axes 3 and 4. The fields are in quadrature but

amplitudes differ.

ANSWER **Elliptical polarization.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q9 Model · syllabus-based

How does Maxwell theory predict the speed of waves in vacuum?

KNOW FIRST

T4 Maxwell's vacuum wave equation P1 School physics bridge M1 Mathematics from the beginning

The source-free curl equations give $\nabla^2 E = \mu_0 \epsilon_0 \partial_t^2 E$. Comparing with the wave equation gives $c = (\mu_0 \epsilon_0)^{-1/2} \simeq 3.00 \times 10^8$ m/s.

ANSWER **Vacuum wave speed is determined by μ_o and ε_o.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q10 Model · syllabus-based

State the assumptions needed for the elementary vacuum EM wave equation.

KNOW FIRST

T4 Maxwell's vacuum wave equation P1 School physics bridge M1 Mathematics from the beginning

The region is vacuum with no free or bound charges/currents, so $\rho=0$, $J=0$, $D=\varepsilon_o E$ and $B=\mu_o H$. The vacuum constants are uniform and the fields

are differentiable. Sources may exist outside the region.

ANSWER **The derivation applies locally in a source-free vacuum region.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q11 Model · syllabus-based

A wave travels along +x and E points +z at an instant. Which way does B point?

KNOW FIRST

T5 Plane-wave geometry and linked fields P1
School physics bridge M1 Mathematics from the beginning

$B = (1/c)\hat{x} \times E$. Since $\hat{x} \times \hat{z} = -\hat{y}$, B points −y. The check is $E \times B = \hat{z} \times (-\hat{y}) = +\hat{x}$.

ANSWER **B points −y.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q12 Model · syllabus-based

A vacuum plane wave has electric peak 300 V/m. Find its magnetic peak and relative phase.

KNOW FIRST

T5 Plane-wave geometry and linked fields P1
School physics bridge M1 Mathematics from the beginning

$B_0 = E_0/c = 300/(3.00 \times 10^8) = 1.00 \times 10^{-6}$ T. E and B reach their signed extrema together for a travelling plane wave; they are in phase.

ANSWER $B_0 = 1.00$ **μT, in phase with E.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q13 Model · syllabus-based

What does transverse mean for a vacuum plane electromagnetic wave?

KNOW FIRST

T5 Plane-wave geometry and linked fields P1 School physics bridge M1 Mathematics from the beginning

Both $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to the propagation vector k: $k \cdot E = k \cdot B = 0$. They are also perpendicular to one another. Their oscillations are not along the travel direction.

ANSWER E, B and k form mutually perpendicular directions.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q14 Model · syllabus-based

Define the vacuum Poynting vector and state its physical meaning and SI unit.

KNOW FIRST

T6 Poynting's theorem P1 School physics bridge M1 Mathematics from the beginning

$\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$. It describes electromagnetic power flow per unit area, directed along energy transport. Its units are W/m².

ANSWER **S is an energy flux density.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q15 Model · syllabus-based

Interpret positive $\mathbf{J} \cdot \mathbf{E}$ in Poynting's theorem.

KNOW FIRST

T6 Poynting's theorem P1 School physics bridge M1 Mathematics from the beginning

Positive $J \cdot E$ means the field does work on charges at that power per unit volume. It removes electromagnetic field energy locally unless replenished by inflow. In an ohmic resistor it equals σE^2.

ANSWER **Positive $J \cdot E$ transfers field energy to matter.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q16 Model · syllabus-based

In a vacuum travelling plane wave, compare electric and magnetic energy densities.

KNOW FIRST

T7 Wave energy and intensity P1 School physics bridge M1 Mathematics from the beginning

$u_E = \epsilon_0 E^2/2$. Using $B = E/c$ and $1/(\mu_0 c^2) = \epsilon_0$ gives $u_B = B^2/(2\mu_0) = u_E$. Equality holds instantaneously for this travelling plane wave.

ANSWER **Each field stores half the total wave energy density.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q17 Model · syllabus-based

Express the average intensity of a vacuum sinusoidal wave using peak and rms electric fields.

KNOW FIRST

T7 Wave energy and intensity P1 School physics bridge M1 Mathematics from the beginning

$I = \frac{1}{2}\epsilon_0 c E_0^2$ and $I = \epsilon_0 c E_{\text{rms}}^2$, with $E_{\text{rms}} = E_0/\sqrt{2}$. The half comes from the cycle average of cosine squared.

ANSWER **Use the half only with peak amplitude.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q18 Model · syllabus-based

What is the pressure on an ideal absorbing surface at normal incidence from a vacuum beam of intensity I?

KNOW FIRST

T8 Momentum and radiation pressure P1 School physics bridge M1 Mathematics from the beginning

An absorbed energy U brings momentum U/c. Per time and area, this gives $p = I/c$. The result assumes complete absorption and normal incidence; its units are N/m².

ANSWER $p = I/c$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q19 Model · syllabus-based

Why does an ideal reflector have twice the normal radiation pressure of an ideal absorber at the same incident intensity?

KNOW FIRST

T8 Momentum and radiation pressure P1 School physics bridge M1 Mathematics from the beginning

An incident momentum component +U/c becomes −U/c after reflection, so the surface gains 2U/c. Absorption gains only U/c. Dividing by area and time gives $p = 2I/c$.

ANSWER **Momentum reversal doubles the transfer.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q20 Model · syllabus-based

A lossless nonmagnetic dielectric has $\epsilon_r = 2.25$. Find refractive index and wave speed.

KNOW FIRST

T9 Waves in linear media P1 School physics bridge M1 Mathematics from the beginning

$n = \sqrt{\mu_r \epsilon_r} = \sqrt{2.25} = 1.5$. Thus $v = c/n = 2.00 \times 10^8$ m/s. This uses a homogeneous nondispersive scalar permittivity and μr=1.

ANSWER **n=1.5 and $v = 2.00 \times 10^8$ m/s.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Section B

Q21 Model · syllabus-based

Show directly that any twice-differentiable $f(z - vt)$ solves the one-dimensional wave equation.

KNOW FIRST

T1 Travelling waves in one dimension P1 School physics bridge M1 Mathematics from the beginning

1. Set $s = z - vt$. Then $\partial_z f = f'(s)$ and $\partial_z^2 f = f''(s)$.
   Why: The derivative of s with respect to z is one.
2. $\partial_t f = -v f'(s)$ and $\partial_t^2 f = v^2 f''(s)$.
   Why: Each time derivative contributes −v.

3. Thus $\partial_t^2 f = v^2 \partial_z^2 f$.
   Why: Both sides equal $v^2 f''(s)$.

4. The shape is unchanged as s remains constant along z=vt+constant.
   Why: This also explains the direction of propagation.

ANSWER **Every smooth profile f(z−vt) is a right-travelling solution.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q22 Model · syllabus-based

Write E for a +z travelling, x-polarized vacuum wave with peak 20 V/m and wavelength 0.60 m, whose field is maximum at z=t=0. Find B.

KNOW FIRST

T2 Sinusoidal waves and phase P1 School physics bridge M1 Mathematics from the beginning

1. $k = 2\pi/0.60 = 10.472$ rad/m and $\omega = ck = 3.142 \times 10^9$ rad/s.
   Why: Vacuum dispersion is $\omega = ck$.

2. Choose cosine with zero phase: $E = 20\cos(kz - \omega t)\hat{x}$.
   Why: Cosine is one at the specified origin.

3. $B = (20/c)\cos(kz - \omega t)\hat{y}$.
   Why: Use the right-handed plane-wave relation.

4. $B_0 = 6.667 \times 10^{-8}$ T.
   Why: Electric and magnetic amplitudes differ by c.

ANSWER $E = 20\cos(10.472z - 3.142 \times 10^9 t)\hat{x}$ **V/m; B has the same phase and +y direction.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q23 Model · syllabus-based

For $E_x = a\cos\phi$ and $E_y = b\cos(\phi + \delta)$, derive the polarization ellipse equation.

KNOW FIRST

T3 Polarization of a wave P1 School physics bridge M1 Mathematics from the beginning

1. Set $X = E_x/a = \cos\phi$ and $Y = E_y/b = \cos\phi\cos\delta - \sin\phi\sin\delta$.
   Why: Expand the cosine of a sum.
2. Rearrange $Y - X\cos\delta = -\sin\phi\sin\delta$.
   Why: Isolate the sine term.
3. Square and use $\sin^2\phi = 1 - X^2$.
   Why: Eliminate the phase.
4. Collect terms: $X^2 + Y^2 - 2XY\cos\delta = \sin^2\delta$.
   Why: This is the general conic relation.
5. For δ=0 or π it degenerates to a line; for equal a,b and δ=±π/2 it is a circle.
   Why: These are limiting phase and amplitude choices.

ANSWER $E_x^2/a^2 + E_y^2/b^2 - 2E_xE_y\cos\delta/(ab) = \sin^2\delta$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q24 Model · syllabus-based

At z=0 take $E_x = E_0 \cos \omega t$, $E_y = E_0 \sin \omega t$. Describe its rotation as viewed from +z looking toward the origin. Avoid an unqualified handedness label.

KNOW FIRST

T3 Polarization of a wave P1 School physics bridge M1 Mathematics from the beginning

1. At t=0 the tip is on +x.
   Why: Cosine is one and sine zero.
2. At T/4 it is on +y, at T/2 on −x and at 3T/4 on −y.
   Why: Advance the phase by π/2 each quarter-cycle.
3. Viewed from +z toward the origin, +x to +y is counterclockwise.
   Why: This is the standard xy-plane orientation from its positive normal.
4. The magnitude is $\sqrt{E_x^2 + E_y^2} = E_0$.
   Why: Sine and cosine squares add to one.

ANSWER Circular motion, counterclockwise from the explicitly stated +z viewpoint.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q25 Model · syllabus-based

Derive the source-free vacuum wave equation for E from Maxwell's equations.

KNOW FIRST

T4 Maxwell's vacuum wave equation P1 School physics bridge M1 Mathematics from the beginning

1. Take curl of $\nabla \times E = -\partial_t B$.
   Why: We want an equation involving E alone.
2. Use $\nabla \times \nabla \times E = \nabla(\nabla \cdot E) - \nabla^2 E = -\nabla^2 E$.
   Why: Source-free electric divergence is zero.
3. The right side becomes $-\partial_t(\nabla \times B) = -\mu_0 \epsilon_0 \partial_t^2 E$.
   Why: Insert Ampère-Maxwell law.
4. Cancel minus signs.
   Why: Both sides carry the same sign.

ANSWER $\nabla^2 E = \mu_0 \epsilon_0 \partial_t^2 E$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q26 Model · syllabus-based

Derive the source-free vacuum wave equation for B, stating the identity and divergence law used.

KNOW FIRST

T4 Maxwell's vacuum wave equation P1 School physics bridge M1 Mathematics from the beginning

1. Curl Ampère-Maxwell: $\nabla \times \nabla \times B = \mu_0 \epsilon_0 \partial_t (\nabla \times E)$.
   Why: This eliminates E after substitution.

2. The left is $-\nabla^2 B$ because $\nabla \cdot B = 0$.
   Why: Use the curl-curl identity.

3. Faraday's law makes the right side $-\mu_0 \epsilon_0 \partial_t^2 B$.
   Why: Its minus sign is essential.

4. Cancel signs and identify speed $c = (\mu_0 \epsilon_0)^{-1/2}$.
   Why: Compare with the standard wave equation.

ANSWER $\nabla^2 B = \mu_0 \epsilon_0 \partial_t^2 B$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q27 Model · syllabus-based

For $E = E_0 \cos(kz - \omega t)\hat{x}$, use Faraday's law to derive B for a vacuum travelling wave.

KNOW FIRST

T5 Plane-wave geometry and linked fields P1 School physics bridge M1 Mathematics from the beginning

1. The y component of curl E is $\partial_z E_x = -kE_0 \sin(kz - \omega t)$.
   Why: Only E_x varies, and only along z.

2. Try $B = B_0 \cos(kz - \omega t)\hat{y}$, so $-\partial_t B_y = -\omega B_0 \sin(kz - \omega t)$.
   Why: The time derivative of cosine contributes +ω before Faraday's minus sign.

3. Equate amplitudes: $kE_0 = \omega B_0$.
   Why: Both sides have the same phase factor.

4. Since $\omega/k = c$, $B_0 = E_0/c$.
   Why: Use vacuum dispersion.

ANSWER **B points +y, is in phase, and has peak E_0/c.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q28 Model · syllabus-based

Someone proposes a +z vacuum wave with E along +x and B along +y but $B_0 = 2E_0/c$. Can the pair satisfy Maxwell's equations? Explain.

KNOW FIRST

T5 Plane-wave geometry and linked fields P1 School physics bridge M1 Mathematics from the beginning

1. Its directions pass the right-hand test, but direction alone is insufficient.
   Why: The fields must satisfy the curl equations.

2. Faraday's law requires $kE_0 = \omega B_0$.
   Why: Derivatives connect the amplitudes.

3. Vacuum wave dispersion requires $\omega = ck$, hence $B_0 = E_0/c$.
   Why: Both conditions must hold together.

4. The proposed factor two violates this unless both amplitudes vanish.
   Why: A nonzero source-free travelling pair has a fixed ratio.

ANSWER **No: the proposed amplitude ratio violates Maxwell's curl equations.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q29 Model · syllabus-based

A region loses field energy at 5 W while charges inside receive 2 W from the field. Find the net outward electromagnetic power.

KNOW FIRST

T6 Poynting's theorem P1 School physics bridge M1 Mathematics from the beginning

1. Integrate Poynting's theorem: $dU/dt + P_{\text{out}} = -P_{\text{matter}}$.
   Why: The divergence theorem converts S to boundary flow.
2. Here $dU/dt = -5$ W and $P_{\text{matter}} = +2$ W.
   Why: Loss of stored energy carries a negative derivative.
3. Thus $-5 + P_{\text{out}} = -2$, giving $P_{\text{out}} = 3$ W.
   Why: Part of the released energy goes to matter and the remainder leaves the region.

ANSWER **Net outward power is 3 W.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q30 Model · syllabus-based

A vacuum plane wave has $E_0 = 100$ V/m. Calculate mean energy density and intensity.

KNOW FIRST

T7 Wave energy and intensity P1 School physics bridge M1 Mathematics from the beginning

1. $\langle u \rangle = \epsilon_0 E_0^2/2 = (8.854 \times 10^{-12})(10^4)/2 = 4.427 \times 10^{-8}$ J/m³.
   Why: The average squared cosine is 1/2.
2. $I = c\langle u \rangle = 13.28$ W/m².
   Why: A travelling wave carries its energy at c.
3. Equivalently $I = E_0^2/(2\eta_0)$ with $\eta_0 \simeq 376.7\ \Omega$.
   Why: This provides an independent ratio check.

ANSWER **Mean energy density 4.427×10^{-8} J/m³; intensity 13.28 W/m².**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q31 Model · syllabus-based

A uniform vacuum beam carries 2 W through area 1.0×10^{-4} m². Find intensity and peak E and B.

KNOW FIRST

T7 Wave energy and intensity P1 School physics bridge M1 Mathematics from the beginning

1. $I = P/A = 2.0 \times 10^4$ W/m².
   Why: Power is uniformly distributed.

2. $E_0 = \sqrt{2I/(\epsilon_0 c)} = 3.881 \times 10^3$ V/m.
   Why: Invert the peak-field intensity relation.

3. $B_0 = E_0/c = 1.294 \times 10^{-5}$ T.
   Why: The wave is in vacuum.

4. The square root is essential because intensity depends on field squared.
   Why: Doubling power does not double field amplitude.

ANSWER **I=2.0×10^4 W/m²; E_0≈3.881 kV/m; B_0≈12.94 μT.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q32 Model · syllabus-based

A 3 W beam hits a surface normally in vacuum. Find force for complete absorption and perfect reflection.

KNOW FIRST

T8 Momentum and radiation pressure P1 School physics bridge M1 Mathematics from the beginning

1. Incident energy per second is P, so incident momentum per second is P/c.
   Why: A vacuum wave has momentum U/c.

2. Absorption gives $F = P/c = 1.0 \times 10^{-8}$ N.
   Why: All incident momentum is retained.

3. Reflection gives $F = 2P/c = 2.0 \times 10^{-8}$ N.
   Why: Normal momentum reverses.

4. Both forces point into the surface along the incoming beam.
   Why: The surface receives forward momentum.

ANSWER **Absorber: 10 nN; reflector: 20 nN.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q33 Model · syllabus-based

At normal incidence a surface reflects fraction R of a vacuum beam's intensity and absorbs the remainder, with no transmission. Derive its pressure.

KNOW FIRST

T8 Momentum and radiation pressure P1 School physics bridge M1 Mathematics from the beginning

1. The absorbed fraction $(1 - R)I$ transfers pressure $(1 - R)I/c$.
   Why: Absorbed momentum is removed from the beam.

2. The reflected fraction RI transfers $2RI/c$.
   Why: Reflection reverses its normal momentum.

3. Add contributions: $p = (1 + R)I/c$.
   Why: Power fractions are mutually exclusive.

4. R=0 and R=1 recover the absorbing and reflecting limits.
   Why: Check the result against known endpoints.

ANSWER $p = (1 + R)I/c$ **for no transmission at normal incidence.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q34 Model · syllabus-based

A lossless homogeneous medium has $\epsilon_r = 4$, $\mu_r = 1$ and electric peak 60 V/m. Find speed, impedance, B_o, H_o and intensity.

KNOW FIRST

T9 Waves in linear media P1 School physics bridge M1 Mathematics from the beginning

1. $v = c/2 = 1.50 \times 10^8$ m/s and $\eta = \eta_0/2 \simeq 188.4$ Ω.
   Why: Use the square-root constitutive relations.

2. $B_0 = E_0/v = 4.0 \times 10^{-7}$ T and $H_0 = E_0/\eta = 0.3185$ A/m.
   Why: The ratios are B/E=1/v and E/H=η.

3. $I = E_0^2/(2\eta) = 9.555$ W/m².
   Why: Average the product of in-phase sinusoidal E and H.

ANSWER **v=1.50×10^8 m/s, η≈188.4 Ω, B_o=0.40 μT, H_o≈0.3185 A/m, I≈9.555 W/m².**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q35 Model · syllabus-based

A 600 nm vacuum wave enters a stationary transparent nonmagnetic medium with n=1.5. Find f, v and wavelength inside. Explain which quantity stays fixed.

KNOW FIRST

T9 Waves in linear media P1 School physics bridge M1 Mathematics from the beginning

1. Vacuum frequency is $f = c/\lambda_0 = 5.0 \times 10^{14}$ Hz.
   Why: Convert nm to metres.

2. Frequency stays the same across a stationary boundary.
   Why: The boundary fields must match at every instant.

3. $v = c/n = 2.0 \times 10^8$ m/s, so $\lambda = v/f = 400$ nm.
   Why: Use the unchanged f with the reduced speed.

ANSWER **f=5.0×10^{14} Hz remains fixed; v=2.0×10^8 m/s; λ=400 nm.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q36 Model · syllabus-based

Show that S=vu for a travelling plane wave in a homogeneous linear lossless nondispersive medium.

KNOW FIRST

T9 Waves in linear media P1 School physics bridge M1 Mathematics from the beginning

1. $u = (\epsilon E^2 + \mu H^2)/2.$
   Why: Both field energies are reversible under the stated assumptions.

2. Since $H = E/\eta$ and $\eta^2 = \mu/\epsilon$, $\mu H^2 = \epsilon E^2$, hence $u = \epsilon E^2$.
   Why: The wave's electric and magnetic energy shares are equal.

3. $S = EH = E^2/\eta.$
   Why: The fields are perpendicular and form a travelling wave.

4. $S/u = 1/(\epsilon\eta) = 1/\sqrt{\mu\epsilon} = v.$
   Why: Substitute the impedance definition.

ANSWER $\mathbf{S} = vu\hat{k}$ **under the stated nondispersive lossless assumptions.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Section C

Q37 Model · syllabus-based

Starting with source-free Maxwell equations, derive the wave speed, transversality and electric/magnetic amplitude relation of a monochromatic plane wave.

KNOW FIRST

T4 Maxwell's vacuum wave equation P1 School physics bridge M1 Mathematics from the beginning

1. Curl Faraday and use the identity $\nabla \times \nabla \times E = \nabla(\nabla \cdot E) - \nabla^2 E$.
   Why: This connects curls to spatial curvature.

2. With zero electric divergence and Ampère-Maxwell law, obtain $\nabla^2 E = \mu_0 \epsilon_0 \partial_t^2 E$.
   Why: The fields are source-free vacuum fields.

3. A cosine of phase $k \cdot r - \omega t$ then requires $\omega^2 = c^2 k^2$, with $c = 1/\sqrt{\mu_0 \epsilon_0}$.
   Why: Second derivatives supply $-k^2$ and $-\omega^2$.

4. The divergence laws give $k \cdot E_0 = k \cdot B_0 = 0$.
   Why: The oscillatory fields have no longitudinal component.

5. Faraday gives $k \times E_0 = \omega B_0$, hence $B_0 = (1/c)\hat{k} \times E_0$.
   Why: The curl law fixes orientation, amplitude and phase.

6. For travel +z choose E along x and B along y with the same cosine.
   Why: Then E cross B points along propagation.

ANSWER **Vacuum waves travel at c and have transverse in-phase fields with $E_0 = cB_0$.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q38 Model · syllabus-based

Derive Poynting's theorem in vacuum with charges and currents, then give its integral interpretation.

KNOW FIRST

T6 Poynting's theorem P1 School physics bridge M1 Mathematics from the beginning

1. Ampère-Maxwell gives $E \cdot (\nabla \times B) = \mu_0 J \cdot E + \mu_0 \epsilon_0 E \cdot \partial_t E$.
   Why: Take its dot product with E.
2. Faraday gives $B \cdot (\nabla \times E) = -B \cdot \partial_t B$.
   Why: Take its dot product with B.
3. Use $\nabla \cdot (E \times B) = B \cdot \nabla \times E - E \cdot \nabla \times B$.
   Why: This combines the curl terms.
4. Divide by μ_0 and identify $u = \epsilon_0 E^2/2 + B^2/(2\mu_0)$ and $S = E \times B/\mu_0$.
   Why: The squared-field derivatives are twice field dotted with its derivative.

5. Obtain $\partial_t u + \nabla \cdot S = -J \cdot E$.
   Why: The remaining term is work on charges.

6. Integrate over volume: $dU/dt + \oint S \cdot da = -\int J \cdot E \, d\tau$.
   Why: The divergence theorem gives outward surface flow.

7. The field's energy loss equals outward radiation plus work delivered to matter.
   Why: This is local and global energy conservation.

ANSWER **Poynting's theorem balances stored energy, boundary energy flux and mechanical/electrical work on charge.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q39 Model · syllabus-based

Derive the vacuum plane-wave energy/intensity relations and use momentum transport to explain radiation pressure for an absorber and reflector.

KNOW FIRST

T8 Momentum and radiation pressure P1 School physics bridge M1 Mathematics from the beginning

1. The plane-wave amplitude relation gives $u_B = u_E = \epsilon_0 E^2/2$.
   Why: Substitute B=E/c into magnetic energy.

2. Total $u = \epsilon_0 E^2$ and $S = cu\hat{k}$.
   Why: Compute the Poynting vector.

3. For sinusoidal peak E_0, $I = \epsilon_0 c E_0^2/2$.
   Why: Average cosine squared.

4. Vacuum momentum density is $g = S/c^2$, so momentum carried by energy U is U/c.
   Why: Divide energy density by c.

5. For normal complete absorption, momentum delivered per time per area gives $p = I/c$.
   Why: Pressure is momentum flux.

6. Perfect reflection reverses the normal momentum and gives $p = 2I/c$.
   Why: The momentum change is twice the incoming value.

7. These formulas refer to vacuum incident waves and ideal stated surface behaviour.
   Why: Inside material, field/matter momentum requires additional care.

ANSWER **Energy and momentum transport jointly predict measurable radiation pressure.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q40 Model · syllabus-based

Derive the wave equation and field ratios in a source-free homogeneous linear lossless nondispersive medium, and identify the limits of the model.

KNOW FIRST

T9 Waves in linear media P1 School physics bridge M1 Mathematics from the beginning

1. Use $D = \epsilon E$, $B = \mu H$, with no free current or charge.
   Why: Constitutive laws specify an isotropic linear medium.

2. Then $\nabla \times E = -\mu \partial_t H$ and $\nabla \times H = \epsilon \partial_t E$.
   Why: The constants are uniform and time independent.

3. Curl the first and use $\nabla \cdot E = 0$ to get $\nabla^2 E = \mu\epsilon \partial_t^2 E$.
   Why: No spatial derivatives of ε or μ appear in a homogeneous region.

4. Thus $v = 1/\sqrt{\mu\epsilon}$ and $n = c/v = \sqrt{\mu_r \epsilon_r}$.
   Why: Compare with the wave equation.

5. A transverse plane wave has $B = \hat{k} \times E/v$, $H = \hat{k} \times E/\eta$, with $\eta = \sqrt{\mu/\epsilon}$.
   Why: The curl equations determine the ratios.

6. $S = E \times H$, and mean intensity is $E_0^2/(2\eta)$.
   Why: Average the in-phase sinusoidal product.

7. Absorbing, conducting, dispersive or anisotropic media need extended equations/constitutive treatment.
   Why: The simple real scalar constants then no longer suffice.

ANSWER **Speed and impedance are separate material properties: v sets B/E, while η sets E/H.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.